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Model Predictive Cabin Climate Control for Autonomous Electric Vehicles: Exploiting Booking, Route, and Battery Forecasts

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Abstract

Autonomous electric vehicles such as robotaxis possess unique forecast capabilities—including passenger booking schedules, planned route profiles, and battery state-of-charge trajectories—that conventional vehicles lack. This paper investigates how model predictive control (MPC) can exploit these forecasts for energy-efficient cabin climate management. A four-node lumped-parameter thermal model comprising the PTC heater element, HVAC duct, cabin air, and interior thermal mass is developed, complemented by a CO_2 mass balance for indoor air quality tracking. The MPC simultaneously optimizes four manipulated variables—heat pump compressor modulation, PTC heater output, blower fan speed, and air recirculation ratio—subject to thermal comfort constraints and a CO_2 limit of 1200 ppm. Three simulation scenarios demonstrate distinct MPC capabilities: (1) predictive pre-conditioning before passenger boarding using booking data achieves 58 % energy savings; (2) anticipation of highway speeds to exploit improved heat exchanger efficiency yields 60 % savings; and (3) SOC-dependent comfort relaxation using a stage parameter that enables graceful degradation at low battery charge results in 74 % savings. All results are benchmarked against a conventional PID baseline controller that lacks forecast access. The findings demonstrate that exploiting the forecast availability inherent to autonomous vehicles enables substantial HVAC energy reductions while maintaining passenger comfort.

Keywords: model predictive control; cabin climate control; autonomous electric vehicles; robotaxi; thermal management; energy efficiency; HVAC; CO_2 tracking; state of charge

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1. Introduction

The transition toward electric mobility is fundamentally reshaping vehicle powertrain and auxiliary system design. Among auxiliary consumers, heating, ventilation, and air conditioning (HVAC) systems represent a particularly significant energy demand. König et al. [1] demonstrated that HVAC energy consumption can exceed that of the powertrain during urban driving cycles at low average speeds, and that battery sizing must account for weather-dependent HVAC loads to achieve target driving ranges. Campisi [2] confirmed this finding using real-world data from multiple European climates, reporting a 44-fold power variation across the temperature range and HVAC contributions of up to 75 % of total power at idle. These results underscore that effective thermal management is not merely a comfort concern but a range-critical design parameter for electric vehicles.

Autonomous electric vehicles—particularly robotaxi fleets operating in urban environments—face a unique tension: passengers expect premium comfort as a service quality indicator,

while fleet operators bear the direct energy cost. Zheng and Mueller [3] provided an overview of the integrated thermal control architecture for the Zoox robotaxi, highlighting the unique challenges of autonomous platforms where no driver is present to report discomfort and compute hardware adds substantial thermal loads. Simultaneously, autonomous vehicles possess forecast capabilities that conventional vehicles lack. The booking system provides advance knowledge of passenger boarding times and occupancy. Route planning yields velocity profiles, heading trajectories, and trip durations. The battery management system (BMS) supplies state-of-charge (SOC) forecasts along the planned route. This comprehensive forecast availability creates an opportunity for predictive control strategies that can anticipate thermal loads, exploit physical couplings, and make energy-aware decisions.

Considerable research has addressed thermal modeling and control for electric vehicles. Pulvirenti et al. [4] developed a lumped-parameter energy balance model for EV cabin HVAC systems and demonstrated that dynamic (unsteady) analysis yields significantly different energy consumption estimates compared to steady-state approaches. Poovendran [5] presented a multi-zone MPC approach for vehicle cabin thermal control with environmental disturbance compensation. In the domain of battery thermal management, Wang et al. [6] proposed both thermal equivalent circuit and artificial neural network models for predicting battery temperature under real-world driving cycles. Kirchgässner et al. [7] introduced thermal neural networks that combine lumped-parameter structure with data-driven parameter identification, achieving high estimation accuracy with physically interpretable states.

Model predictive control has emerged as a promising paradigm for coordinated thermal management. Wang et al. [8] demonstrated one of the earliest applications of nonlinear MPC to automotive air conditioning in connected and automated vehicles, achieving up to 9 % energy efficiency improvement by exploiting vehicle speed preview. Amini et al. [9] extended this to a hierarchical MPC framework for coordinated cabin and battery thermal management, addressing multi-timescale dynamics through robust constraint handling under speed prediction uncertainty. Glos et al. [10] applied nonlinear MPC to simultaneously control cabin temperature and air quality in fully electric vehicles, implemented on an automotive-grade Infineon AURIX microcontroller, demonstrating embedded real-time feasibility. Liu and Zhang [11] proposed an MPC-based strategy to simultaneously control the battery thermal management system and cabin air conditioning, demonstrating a 4.3 % reduction in recharging energy and 6.5 % improvement in overall energy consumption. Schutzeich et al. [12] applied MPC to cabin conditioning of battery electric vehicles with explicit CO_2 tracking and recirculation control, reporting energy savings of up to 15.4 % in cold and 39.7 % in hot conditions compared to a rule-based strategy. Fussey et al. [13] demonstrated adaptive MPC on a Jaguar I-PACE on a climatic chassis dynamometer, optimizing air recirculation to reduce heating energy while maintaining comfort—one of the few real-vehicle MPC demonstrations. Nam and Ahn [14] combined an artificial neural network surrogate model with an infinity-horizon MPC framework for battery thermal management, achieving energy reductions of 27.8 % to 78.9 % depending on ambient temperature. Padisala and Dey [15] addressed the specific challenge of autonomous electric vehicles under cold conditions, using MPC to dynamically allocate battery power between propulsion, HVAC, and battery preconditioning.

Two cross-cutting themes merit particular attention. First, cabin air quality and energy consumption are fundamentally coupled through the recirculation ratio. Wei et al. [16] systematically quantified this trade-off, showing that dynamic recirculation control can reduce climate system power by up to 27 % while maintaining CO_2 below 1000 ppm, with occupancy-dependent limits on the maximum recirculation ratio. Lesage et al. [17] optimized blower flow rate and recirculation ratio jointly for air quality and energy, reporting

range gains of 9 km to 26 km compared to full fresh-air operation. Second, the comfort–range trade-off has been formalized by Lahlou et al. [18], who used dynamic programming to optimize thermal comfort while preserving driving range, modeling the vehicle together with the passengers' thermo-physiological behavior.

Despite these advances, a gap remains in exploiting the *full* forecast availability unique to autonomous vehicles. Existing MPC approaches for cabin climate control typically use weather forecasts or simplified disturbance models, but do not leverage the detailed booking, route, and battery information that a robotaxi's fleet management system provides. Furthermore, the concept of SOC-dependent comfort relaxation—consciously trading thermal comfort for range preservation when battery charge is critically low—has not been systematically investigated as an MPC design parameter.

This paper makes the following contributions:

1. A four-node lumped-parameter thermal cabin model with CO_2 mass balance, incorporating heat pump partial-load efficiency, speed-dependent heat exchanger performance, and blower–recirculation coupling, suitable for real-time MPC deployment.
2. An MPC formulation that exploits three distinct robotaxi-specific forecast types—passenger schedule, velocity profile, and SOC trajectory—through measured disturbances and a novel SOC stage parameter.
3. Systematic demonstration of forecast-driven energy savings (58 % to 74 %) across three scenarios, each isolating a different MPC capability, benchmarked against a conventional PID controller.

The remainder of this paper is organized as follows. Section 2 presents the thermal cabin model and HVAC system components. Section 3 describes the MPC formulation and PID baseline. Section 4 presents three simulation scenarios with quantitative comparisons. Section 5 discusses the findings, limitations, and practical implications. Section 6 summarizes the key conclusions.

2. System Model

2.1. Autonomous Vehicle Cabin Architecture

The system under consideration is an autonomous electric vehicle (robotaxi) with an integrated thermal management system comprising multiple hydraulic circuits. Figure 1 shows the complete hydraulic architecture, which includes a reversible heat pump, a high-voltage PTC heater, a front radiator, the cabin HVAC unit, and dedicated cooling circuits for the power electronics (PEEM) and traction battery. The two coolant loops (heating circuit in red, cooling circuit in blue) are interconnected through the heat pump, which can transfer thermal energy between circuits depending on the operating mode.

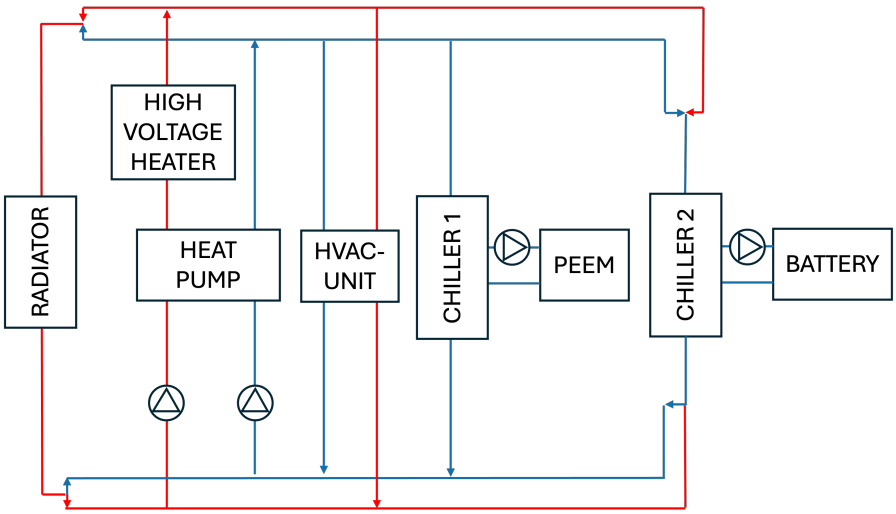

Figure 1. Hydraulic schematic of the vehicle thermal management system. The heating circuit (red) connects the radiator, high-voltage heater, heat pump, and HVAC unit. The cooling circuit (blue) serves the power electronics (PEEM) and battery via dedicated chillers. This work focuses on the cabin conditioning path (heat pump, PTC heater, HVAC unit, and radiator); battery and PEEM thermal management are outside the model scope.

This work focuses on the cabin conditioning subsystem: the heat pump with speed-dependent radiator efficiency, the PTC auxiliary heater for extreme cold conditions, the variable-speed blower fan, and the controllable fresh air/recirculation flap. Battery and power electronics thermal management are not modeled explicitly; the battery state of charge (SOC) enters the control formulation as a stage parameter (see Section 3.1.2).

The restriction to cabin conditioning is a deliberate architectural choice. Safety-critical subsystems—battery thermal management, power electronics cooling, and autonomous driving compute—are excluded from the MPC optimization loop. This separation is motivated by the intended progression toward data-driven (neural network) prediction models in future work. While a neural network predictor that temporarily misjudges the cabin temperature produces a minor comfort deviation, analogous prediction errors in battery thermal management could lead to thermal runaway, and in drive unit cooling to component degradation. By confining the MPC to the non-safety-critical cabin domain, the eventual transition from physics-based to learned models becomes viable without requiring safety certification of the neural network itself—a significant barrier in the automotive industry.

The fleet management system provides the MPC with measured disturbances that are known over the prediction horizon: ambient temperature T_{amb}, solar irradiance I_{solar}, vehicle heading ψ, number of passengers n_{pass}, and vehicle speed v. Additionally, the SOC trajectory is available as a stage parameter that influences the optimization objective rather than the plant dynamics.

2.2. Lumped-Parameter Thermal Model

A four-node lumped-parameter model captures the thermal dynamics of the cabin system. The nodes represent (1) the PTC heater element, (2) the HVAC duct, (3) the cabin air volume, and (4) the interior thermal mass (dashboard, seats, trim). Figure 2 illustrates the thermal network and heat flow paths.

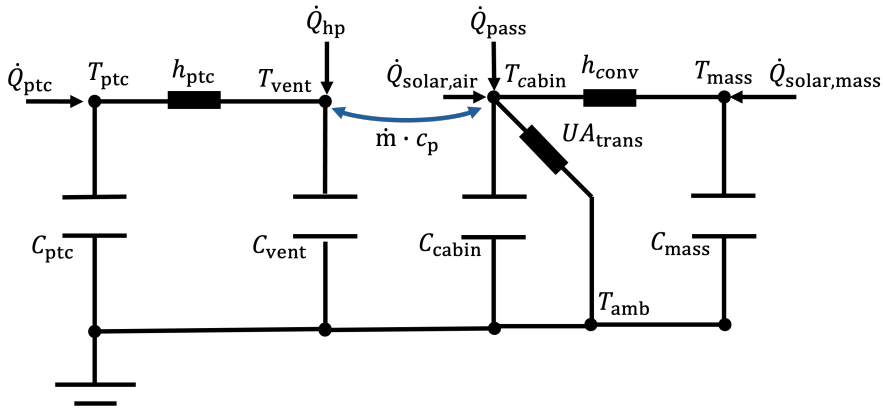

Figure 2. Four-node thermal network model. Heat flows include the heat pump output $\dot{Q}_{hp}$, PTC heater $\dot{Q}_{ptc}$, transmission losses $\dot{Q}_{trans}$, solar gains $\dot{Q}_{solar}$, passenger heat $\dot{Q}_{pass}$, and convective exchange between cabin air and interior mass. The thermal mass node T_{mass} is not directly measurable and therefore hidden from the MPC.

2.2.1. PTC Heater Element

The PTC element has a small thermal capacitance and fast response:

$$C_{ptc}\frac{dT_{ptc}}{dt} = u_{ptc} \cdot \dot{Q}_{ptc,max} - h_{ptc}(T_{ptc} - T_{vent}), \tag{1}$$

where $C_{ptc} = 1500\,\mathrm{J/K}$ is the thermal capacitance, $h_{ptc} = 200\,\mathrm{W/K}$ is the heat transfer coefficient to the duct air, and $\dot{Q}_{ptc,max} = 6000\,\mathrm{W}$ is the maximum PTC power. The PTC operates only when $T_{amb} < -5\,^{\circ}\mathrm{C}$; it is inactive in all cooling scenarios presented in this work.

2.2.2. HVAC Duct

The duct node receives heat from both the heat pump and PTC element and transfers it to the mixed inlet air:

$$C_{vent}\frac{dT_{vent}}{dt} = \dot{Q}_{hp} + h_{ptc}(T_{ptc} - T_{vent}) - \dot{m}\,c_p(T_{vent} - T_{inlet}), \tag{2}$$

with $C_{vent} = 5000\,\mathrm{J/K}$ and the blower mass flow rate $\dot{m} = \dot{m}_{max} \cdot u_{blower}$. The inlet air temperature is a mixture of fresh and recirculated air:

$$T_{inlet} = f_{fresh} \cdot T_{amb} + (1 - f_{fresh}) \cdot T_{cabin}, \tag{3}$$

where the fresh air fraction is $f_{fresh} = \max(f_{min},\ 1 - u_{recirc})$ with a minimum of $f_{min} = 0.1$ to model imperfect cabin sealing.

2.2.3. Cabin Air

The cabin air node is the primary controlled variable:

$$C_{cabin}\frac{dT_{cabin}}{dt} = \dot{m}\,c_p(T_{vent} - T_{cabin}) + \dot{Q}_{trans} + \dot{Q}_{solar,air} + \dot{Q}_{pass} + h_{conv}A_{int}(T_{mass} - T_{cabin}), \tag{4}$$

with $C_{cabin} = 50\,000\,\mathrm{J/K}$. The individual heat flow contributions are:

$$\dot{Q}_{trans} = (UA_{opaque} + UA_{window})(T_{amb} - T_{cabin}), \tag{5}$$

$$\dot{Q}_{solar,air} = 0.3 \cdot A_{window} \cdot \tau_{window} \cdot I_{solar} \cdot f(\psi), \tag{6}$$

$$\dot{Q}_{pass} = n_{pass} \cdot 90\,\mathrm{W}. \tag{7}$$

The solar gain is split with 30 % absorbed by the cabin air and 70 % by the interior thermal mass, and $f(\psi)$ is a heading-dependent angle factor.

2.2.4. Interior Thermal Mass

The interior mass (dashboard, seats, trim panels) acts as a thermal buffer with large capacitance:

$$C_{\text{mass}} \frac{dT_{\text{mass}}}{dt} = \dot{Q}_{\text{solar,mass}} + h_{\text{conv}} A_{\text{int}} (T_{\text{cabin}} - T_{\text{mass}}), \tag{8}$$

with $C_{\text{mass}} = 120\,000\,\text{J/K}$ and $\dot{Q}_{\text{solar,mass}} = 0.7 \cdot A_{\text{window}} \cdot \tau_{\text{window}} \cdot I_{\text{solar}} \cdot f(\psi)$. Critically, T_{mass} is *not directly measurable* in a real vehicle and is therefore hidden from the MPC, which must infer its influence through the observable cabin air temperature.

2.3. CO_2 Mass Balance

Indoor air quality is tracked via the cabin CO_2 concentration:

$$V_{\text{cabin}} \frac{dC_{CO_2}}{dt} = n_{\text{pass}} \cdot R_{CO_2} - \frac{\dot{m}_{\text{fresh}}}{\rho_{\text{air}}} (C_{CO_2} - C_{CO_2,\text{amb}}), \tag{9}$$

where $\dot{m}_{\text{fresh}} = \dot{m}_{\text{max}} \cdot u_{\text{blower}} \cdot f_{\text{fresh}}$. This equation couples CO_2 dilution to both blower speed and recirculation ratio, creating a three-way trade-off: increasing recirculation saves thermal energy (less outside air to condition) but degrades air quality, while increasing the blower improves both ventilation and HVAC heat transfer but increases fan power.

2.4. HVAC System Components

2.4.1. Heat Pump with Partial-Load Efficiency

The reversible heat pump provides the primary thermal output. Its effective COP improves at partial load due to reduced compressor losses:

$$\text{COP}_{\text{eff}} = \text{COP}_{\text{base}}(T_{\text{amb}}) \cdot (1 + \alpha_{\text{PLR}} \cdot (1 - u_{\text{hvac}})), \tag{10}$$

with the partial-load ratio factor $\alpha_{\text{PLR}} = 0.3$. The base COP depends on ambient temperature and ranges from 1.5 to 4.0 in cooling mode.

2.4.2. Speed-Dependent Heat Exchanger Efficiency

The front heat exchanger (condenser in cooling mode) relies on airflow for heat rejection. Its efficiency depends on vehicle speed:

$$\eta_{\text{rad}}(v) = 0.5 + 0.5 \cdot \left(1 - e^{-v/15}\right). \tag{11}$$

At standstill ($v = 0$), only the electric fan provides airflow ($\eta_{\text{rad}} = 0.50$). At highway speed ($v = 25\,\text{m/s}$), ram air increases efficiency to $\eta_{\text{rad}} = 0.91$. This speed dependence creates an opportunity for MPC to time compressor operation with high-speed driving segments.

The effective heat pump output delivered to the cabin is:

$$\dot{Q}_{\text{hp}} = u_{\text{blower}} \cdot u_{\text{hvac}} \cdot \dot{Q}_{\text{hp,max}} \cdot \eta_{\text{rad}}(v), \tag{12}$$

where $\dot{Q}_{\text{hp,max}} = 5000\,\text{W}$ in cooling mode. The blower gating factor u_{blower} reflects the physical reality that conditioned air must be actively blown into the cabin.

Table 1 summarizes all model parameters.

Table 1. Thermal model and HVAC system parameters.

Parameter	Symbol	Value	Unit
Cabin air thermal capacitance	C_{cabin}	50,000	J/K
Interior mass capacitance	C_{mass}	120,000	J/K
HVAC duct capacitance	C_{vent}	5,000	J/K
PTC element capacitance	C_{ptc}	1,500	J/K
Opaque envelope UA-value	UA_{opaque}	15.0	W/K
Window UA-value	UA_{window}	12.5	W/K
Window area	A_{window}	2.5	m^2
Window transmittance	τ_{window}	0.6	–
Interior surface area	A_{int}	8.0	m^2
Interior convective HTC	h_{conv}	10.0	$W/(m^2\,K)$
PTC–duct HTC	h_{ptc}	200	W/K
Max. blower mass flow	$\dot{m}_{\text{max}}$	0.08	kg/s
Min. fresh air fraction	f_{min}	0.1	–
Cabin volume	V_{cabin}	3.0	m^3
CO_2 gen. per person	R_{CO_2}	5×10^{-6}	m^3/s
Ambient CO_2	$C_{CO_2,\text{amb}}$	420	ppm
Max. HP cooling power	$\dot{Q}_{\text{hp,max}}$	5,000	W
Max. PTC power	$\dot{Q}_{\text{ptc,max}}$	6,000	W
PLR correction factor	α_{PLR}	0.3	–

3. Control Design

Figure 3 illustrates the MPC control architecture. The prediction model—comprising the four WhiteBox ODEs from Sections 2.2 and 2.3 and the HVAC component models from Section 2.4—is embedded within the optimizer (IPOPT), which solves a nonlinear program at each time step. Sensor measurements of T_{ptc}, T_{vent}, T_{cabin}, and C_{CO_2} initialize the prediction at each step; no state estimator is required since these states are directly measurable. The hidden state T_{mass} is not available to the controller.

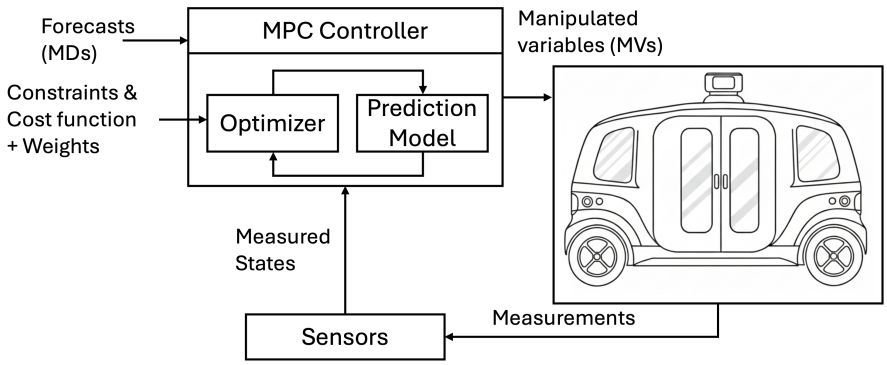

Figure 3. MPC control architecture for autonomous vehicle cabin conditioning. The optimizer receives forecast trajectories of measured disturbances (MDs) from the fleet management system and sensor measurements from the vehicle. The cost function weights are modulated by the SOC stage parameter. The optimizer computes optimal trajectories for all four manipulated variables (MVs) over the prediction horizon.

The three signal categories shown in Figure 3 are:

- **Forecasts (MDs)**: T_{amb} and I_{solar} from weather services, ψ (heading) and v (velocity) from route planning, and n_{pass} from the booking system. These enter the prediction model as known disturbance trajectories over the full horizon.

- **Manipulated variables (MVs)**: u_{hvac} (heat pump), u_{ptc} (PTC heater), u_{blower} (fan speed), and u_{recirc} (recirculation flap). The optimizer determines optimal trajectories; only the first step is applied (receding horizon principle).
- **Constraints & weights**: The cost function weights w_T, w_{CO_2}, w_E, and $w_{\Delta u}$ define the trade-off between comfort, air quality, energy, and actuator smoothness. The comfort weight w_T is modulated by the SOC stage parameter (Section 3.1.2) to enable energy-aware operation at low battery charge.

3.1. MPC Formulation

The MPC optimizes four manipulated variables (MVs) at each time step k over a receding horizon of $N = 20$ steps (20 min at $\Delta t = 60$ s):

- $u_{\text{hvac}} \in [0, 1]$: heat pump compressor modulation,
- $u_{\text{ptc}} \in [0, 1]$: PTC heater modulation,
- $u_{\text{blower}} \in [0.1, 1]$: blower fan speed,
- $u_{\text{recirc}} \in [0, 1]$: recirculation ratio (0 = full fresh air).

3.1.1. Objective Function

The MPC minimizes a multi-objective cost function that balances thermal comfort, air quality, energy consumption, and actuator smoothness:

$$J = \sum_{k=0}^{N-1} \Big[\underbrace{w_T (1 + \alpha_{\text{pass}}\, n_k)\, \varphi(T_k)}_{\text{comfort}} + \underbrace{w_{\text{CO}_2}\, \max(0,\, C_k - 800)^2}_{\text{air quality}} + \underbrace{w_E\, \hat{P}_k^2}_{\text{energy}} + \underbrace{\sum_j w_{\Delta u_j}\, \Delta u_{j,k}^2}_{\text{smoothness}} \Big], \tag{13}$$

where the comfort penalty uses a dead-band formulation:

$$\varphi(T) = \max(0,\ T - T_{\text{ub}})^2 + \max(0,\ T_{\text{lb}} - T)^2, \tag{14}$$

with comfort band boundaries $T_{\text{lb}} = 20\,^\circ\text{C}$ and $T_{\text{ub}} = 24\,^\circ\text{C}$. No penalty is applied within the comfort band, allowing the MPC to freely optimize for energy.

The passenger-dependent weighting factor $(1 + \alpha_{\text{pass}}\, n_k)$ with $\alpha_{\text{pass}} = 0.5$ scales the comfort priority with occupancy. Note that the physical effect of passengers—metabolic heat load $\dot{Q}_{\text{pass}} = n \times 90$ W and CO_2 generation—is already captured in the state equations (Equations (4) and (9)). The weighting factor is an *additional* design choice reflecting fleet operator priorities: a ride with four passengers represents greater aggregate discomfort risk and higher reputational exposure than a single-occupant trip. In a robotaxi context, each passenger is a potential source of service ratings, making comfort violations during high-occupancy rides disproportionately costly for the operator. Conversely, when the vehicle is empty ($n = 0$), the effective comfort weight drops to its base value, allowing the MPC to prioritize energy efficiency during repositioning or pre-conditioning phases.

The energy term uses an approximation of electrical power consumption:

$$\hat{P}_k = u_{\text{hvac},k} \cdot \frac{\dot{Q}_{\text{hp,max}}}{\text{COP}_{\text{nom}}} + u_{\text{ptc},k} \cdot \dot{Q}_{\text{ptc,max}}, \tag{15}$$

which inherently penalizes PTC usage (COP $= 1.0$) more heavily than heat pump operation (COP ≈ 2.5).

3.1.2. SOC Stage Parameter

A distinctive feature of the proposed MPC is the use of battery SOC as a *stage parameter*—a time-varying quantity that modifies the cost function rather than the plant dynamics.

When the SOC drops below a critical threshold ($\mathrm{SOC}_{\mathrm{thr}} = 0.1$), the comfort weight is reduced:

$$w_T(\mathrm{SOC}_k) = \begin{cases} w_{T,\mathrm{base}} & \text{if } \mathrm{SOC}_k \geq 0.1, \\ 0.2 \cdot w_{T,\mathrm{base}} & \text{if } \mathrm{SOC}_k < 0.1. \end{cases} \tag{16}$$

This mechanism enables *graceful degradation*: the MPC consciously trades thermal comfort for range preservation when the battery is critically low. Since the MPC has a 20 min prediction horizon, it anticipates the SOC threshold crossing and begins transitioning smoothly before the threshold is actually reached.

The key distinction is: measured disturbances (e.g., T_{amb}, n_{pass}, v) affect the *plant dynamics* (state equations), whereas the SOC stage parameter affects the *optimization objective* (cost function weights). SOC does not appear in any thermal ODE; instead, the HVAC power consumption reduces SOC, but this reverse causality is outside the thermal model scope.

3.1.3. Constraints

The optimization is subject to:

$$0\,\mathrm{ppm} \leq C_{\mathrm{CO_2}} \leq 1200\,\mathrm{ppm} \qquad \text{(Pettenkofer air quality limit)}, \tag{17}$$

$$0 \leq u_{\mathrm{hvac}} \leq 1, \qquad 0 \leq u_{\mathrm{ptc}} \leq 1,$$
$$0.1 \leq u_{\mathrm{blower}} \leq 1, \qquad 0 \leq u_{\mathrm{recirc}} \leq 1. \tag{18}$$

The minimum blower constraint of 0.1 ensures continuous ventilation. The nonlinear program is solved using IPOPT with CasADi for automatic differentiation of the physics-based predictor models. The Data-Driven Model Predictive Control (DDMPC) framework [19] provides the software infrastructure for integrating the WhiteBox predictors with the NLP solver. Table 2 summarizes the MPC tuning parameters.

Table 2. MPC objective function weights and configuration.

Parameter	Value	Description
w_T	100–5000	Temperature band violation (scenario-dependent)
α_{pass}	0.5	Passenger comfort scaling factor
$w_{\mathrm{CO_2}}$	30–50	CO_2 above 800 ppm
w_E	10^{-4}	Electrical energy approximation
$w_{\Delta u_{\mathrm{hvac}}}$	10	HVAC smoothness
$w_{\Delta u_{\mathrm{blower}}}$	5	Blower smoothness
$w_{\Delta u_{\mathrm{recirc}}}$	5	Recirculation smoothness
N	20	Prediction horizon (steps)
Δt	60 s	Sample time
Solver	IPOPT	Nonlinear programming solver

3.2. PID Baseline Controller

To quantify the benefit of forecast exploitation, a conventional PID controller serves as the baseline. The PID controls u_{hvac} based on the cabin temperature error, with a separate proportional–integral (PI) controller for the blower speed u_{blower}. The recirculation ratio is fixed at $u_{\mathrm{recirc}} = 0.5$. Critically, the PID has no access to any forecast information—it reacts purely to the current temperature deviation. It cannot anticipate passenger boarding, exploit upcoming speed changes, or adapt to battery state. This makes it a representative benchmark for conventional (non-predictive) cabin climate control.

4. Results

Three simulation scenarios are designed to isolate and demonstrate distinct MPC capabilities. All scenarios operate in cooling mode with the PTC heater inactive. Table 3 summarizes the scenario configurations.

Table 3. Simulation scenario configurations.

	S1: Pre-Cond.	**S2: Highway**	**S3: SOC**
Duration	30 min	30 min	120 min
$T_{\text{cabin},0}$	28 °C	30 °C	22 °C
$T_{\text{mass},0}$	30 °C	35 °C	24 °C
T_{amb}	32 °C	33 °C	32 °C
I_{solar}	800 W/m^2	600 W/m^2	600 W/m^2
Passengers	$0 \rightarrow 3$ at t=8 min	2 (constant)	3 (constant)
Velocity	0 m/s (parked)	$5 \rightarrow 25$ m/s	8 m/s (city)
SOC profile	–	–	20% $\rightarrow$ 0% linear
Forecast exploited	n_{pass}	v	SOC

4.1. Scenario 1: Pre-Conditioning Before Passenger Boarding

This scenario demonstrates the primary advantage of robotaxi MPC: predictive cabin preparation using booking data. The cabin starts at 28 °C (moderately warm after short parking). Three passengers board at $t = 8$ min and alight at $t = 25$ min. The MPC receives the passenger schedule as a forecast and can prepare the cabin accordingly.

Figure 4 compares the MPC and PID responses. The PID immediately applies maximum compressor output ($u_{\text{hvac}} = 1.0$) to cool the cabin, which is energetically inefficient due to the low partial-load COP at full power. The MPC, by contrast, uses a gradual compressor ramp ($u_{\text{hvac}} \approx 0.4$–$0.8$), deliberately operating at partial load where the COP is higher (Equation (10)). Both controllers achieve comparable cabin temperatures at the boarding time ($t = 8$ min): PID reaches 22.4 °C, MPC reaches 23.1 °C—both within the comfort band.

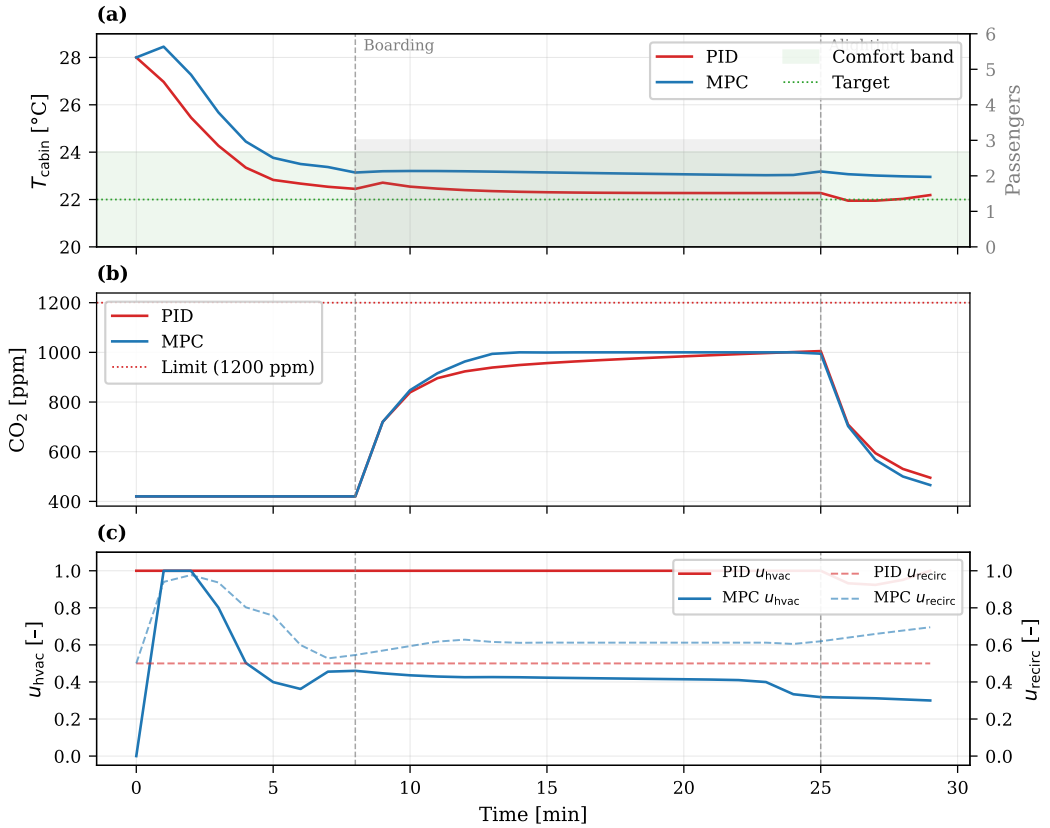

Figure 4. Scenario 1: Pre-conditioning before passenger boarding. (**a**) Cabin temperature and passenger count. (**b**) CO_2 concentration. (**c**) Heat pump compressor modulation u_{hvac} (solid) and recirculation u_{recirc} (dashed). Dashed vertical lines indicate boarding ($t = 8$ min) and alighting ($t = 25$ min).

The recirculation control in Figure 4c further illustrates the difference: the PID keeps u_{recirc} fixed at 0.5 throughout, whereas the MPC actively modulates recirculation—increasing it during the initial cool-down to reduce the thermal load from fresh air, then reducing it before passenger boarding to pre-ventilate the cabin and lower CO_2 levels.

Table 4 summarizes the quantitative results. The MPC achieves 58 % total energy savings while maintaining comparable comfort and air quality. The key mechanism is partial-load efficiency: the MPC operates the compressor at moderate output for longer, rather than at maximum power briefly.

Table 4. Scenario 1 results: Pre-conditioning comparison.

Metric	PID	MPC	Improvement
T_{cabin} at boarding	22.4 °C	23.1 °C	–
Energy total	367 Wh	154 Wh	−58 %
Energy pre-cond. ($t < 8$ min)	76 Wh	54 Wh	−29 %
CO_2 maximum	1005 ppm	1000 ppm	similar

Notably, the same anticipation mechanism works in reverse. When the MPC forecasts that passengers will alight (internal heat load dropping from $n \times 90$ W to zero), it reduces the cooling effort several minutes before the actual departure. The already-conditioned cabin serves as a thermal buffer during this ramp-down, and the compressor transitions to an even more efficient partial-load operating point rather than maintaining high output until the load step occurs. This bidirectional exploitation of the passenger forecast—pre-

conditioning *before* boarding and pre-relaxation *before* alighting—is a direct consequence of the MPC's ability to plan ahead over the full prediction horizon.

4.2. Scenario 2: Highway Speed Anticipation

This scenario demonstrates MPC's ability to exploit the velocity forecast for improved heat exchanger efficiency. The vehicle starts in city traffic ($v = 5\,\mathrm{m/s}$, $\eta_{\mathrm{rad}} = 0.64$) and transitions to highway driving ($v = 25\,\mathrm{m/s}$, $\eta_{\mathrm{rad}} = 0.91$) at $t \approx 5\,\mathrm{min}$. Two passengers are present throughout. The cabin starts at 30 °C with the interior mass at 35 °C (recently parked in sun).

The MPC, knowing the upcoming speed increase, can strategically time its cooling effort. At highway speed, each unit of compressor power produces 42 % more cooling effect than at city speed ($\eta_{\mathrm{rad}} = 0.91$ vs. 0.64). Figure 5 shows that the MPC settles at 23.5 °C rather than 22 °C, operating within the comfort band while accepting a 1.3 K offset to reduce energy consumption significantly. The compressor modulation in Figure 5c highlights this strategy: the PID maintains high u_{hvac} throughout, while the MPC reduces compressor effort once highway speed is reached and the improved radiator efficiency allows more effective cooling per unit of electrical input. The resulting cumulative energy diverges rapidly during the highway phase.

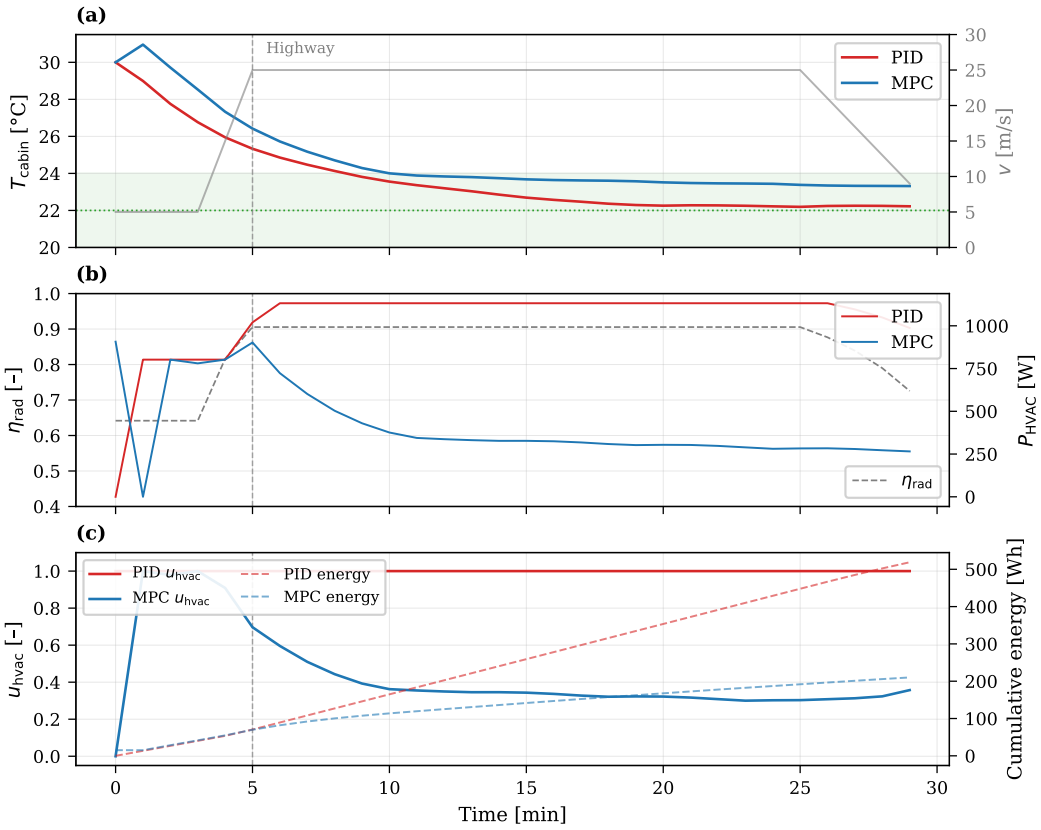

Figure 5. Scenario 2: Highway speed anticipation. (**a**) Cabin temperature and vehicle speed. (**b**) Radiator efficiency $\eta_{\mathrm{rad}}(v)$ and HVAC electrical power. (**c**) Heat pump modulation u_{hvac} (solid) and cumulative energy consumption (dashed). The dashed vertical line indicates the city-to-highway transition.

Table 5. Scenario 2 results: Highway speed anticipation.

Metric	PID	MPC	Improvement
Energy total	519 Wh	210 Wh	−60 %
Energy highway phase	375 Wh	132 Wh	−65 %
T_{cabin} steady state	22.2 °C	23.5 °C	both in band
η_{rad} city / highway	0.64 / 0.91	0.64 / 0.91	identical

4.3. Scenario 3: SOC-Dependent Comfort Relaxation

This scenario demonstrates the most advanced MPC capability: *graceful degradation* through SOC-aware comfort management. The cabin starts at the target temperature (22 °C) with three passengers. The SOC decreases linearly from 20 % to 0 % over 2 h, simulating a low-battery situation where the vehicle must reach a charger. The SOC threshold for comfort relaxation is set at 10 %.

Figure 6 reveals a fundamental behavioral difference. The PID maintains 22 °C at near-maximum compressor effort ($\bar{u}_{\text{hvac}} = 0.98$) throughout, consuming 1703 Wh regardless of battery state. The MPC operates energy-efficiently from the start ($\bar{u}_{\text{hvac}} = 0.29$), allowing the cabin temperature to settle near 24 °C—the upper edge of the comfort band. When the MPC's prediction horizon detects the upcoming SOC threshold at approximately $t = 41$ min (SOC ≈ 13 %), it begins reducing cooling effort further. After the threshold crossing at $t = 60$ min (SOC $= 10$ %), the cabin temperature gradually rises to 25.2 °C as comfort weight is reduced per Equation (16).

The coordinated multi-actuator strategy is visible in Figure 6b–c. The MPC reduces u_{blower} together with u_{hvac} to lower overall HVAC power, while simultaneously increasing u_{recirc} to reduce the thermal load from fresh air intake. The PID, by contrast, keeps u_{recirc} fixed at 0.5 and cannot coordinate these actuators. This multi-variable coordination is a key advantage of MPC and directly contributes to the energy savings shown in the cumulative energy curves (Figure 6c).

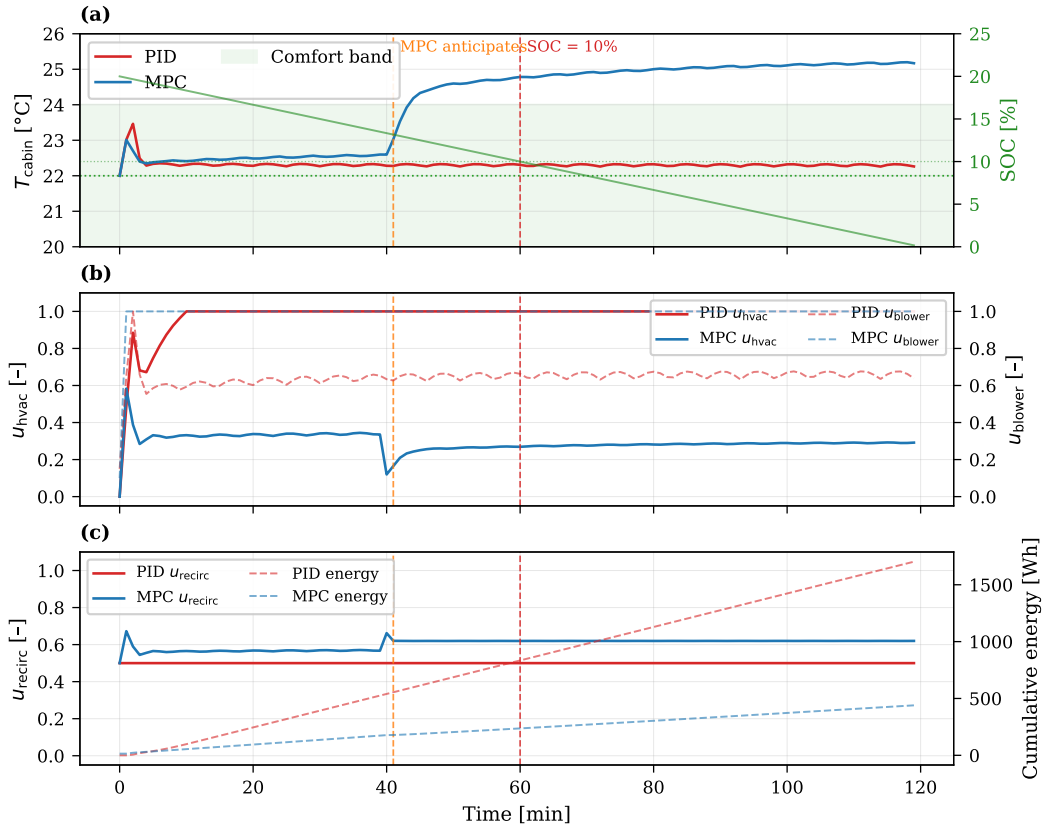

Figure 6. Scenario 3: SOC-dependent comfort relaxation. (**a**) Cabin temperature, comfort band, and SOC profile. (**b**) Heat pump modulation u_{hvac} (solid) and blower modulation u_{blower} (dashed). (**c**) Recirculation u_{recirc} (solid) and cumulative energy consumption (dashed). The MPC anticipates the SOC threshold ($t \approx 41$ min, orange marker) and begins transitioning before the actual threshold crossing ($t = 60$ min, red dashed line).

Table 6. Scenario 3 results: SOC-dependent comfort relaxation.

Metric	PID	MPC	Improvement
Energy total	1703 Wh	440 Wh	−74 %
Energy after threshold	883 Wh	206 Wh	−77 %
Mean u_{hvac}	0.98	0.29	−70 %
T_{cabin} mean	22.3 °C	24.1 °C	MPC allows drift
T_{cabin} final	22.3 °C	25.2 °C	graceful degradation
Comfort violation	0 K · min	71 K · min	accepted trade-off

The comfort violation metric (71 K · min) quantifies the conscious trade-off: the MPC accepts minor comfort deviations to preserve 74 % of the energy that the PID would have consumed. In a real low-battery scenario, this energy difference may determine whether the vehicle reaches the charger.

4.4. Comparative Summary

Figure 7 and Table 7 present the overall comparison across all scenarios. The MPC consistently achieves substantial energy savings by exploiting different forecast types.

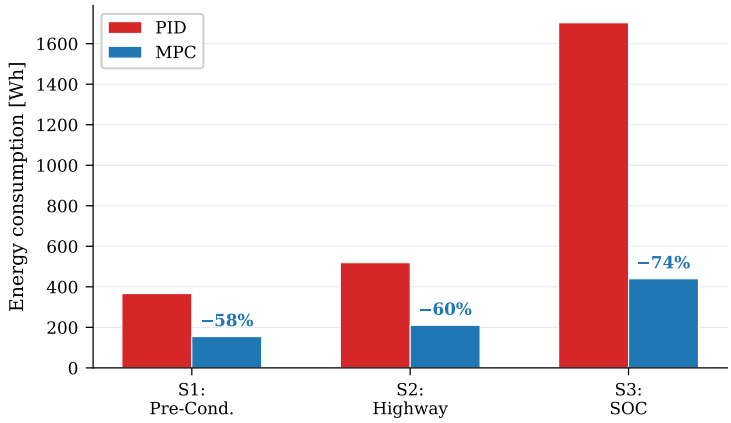

Figure 7. Energy consumption comparison across all three scenarios. The MPC achieves 58 % to 74 % savings compared to the PID baseline.

Table 7. Summary of energy savings across all scenarios.

Scenario	Forecast Used	PID [Wh]	MPC [Wh]	Savings
S1: Pre-Conditioning	n_{pass}	367	154	58 %
S2: Highway Anticipation	v	519	210	60 %
S3: SOC Relaxation	SOC (stage param.)	1,703	440	74 %

Each scenario isolates a different MPC mechanism:

- **S1** demonstrates *anticipation*: the MPC uses the booking forecast to prepare the cabin before passengers board, exploiting partial-load COP efficiency.
- **S2** demonstrates *physical coupling exploitation*: the MPC times compressor operation with highway speeds where the heat exchanger is most effective.
- **S3** demonstrates *energy-aware operation*: the MPC uses the SOC stage parameter to consciously trade comfort for range when the battery is critically low.

5. Discussion

5.1. Comparison with Literature

The energy savings reported in this work (58 % to 74 %) are considerably higher than those reported by Schutzeich et al. [12] (15.4 % to 39.7 %) and Liu and Zhang [11] (4.3 %–6.5 %). Several factors explain this difference. First, the baseline comparison differs: this work compares against a reactive PID controller, whereas Schutzeich et al. used an already-optimized rule-based strategy. Second, the proposed MPC exploits robotaxi-specific forecasts (passenger schedule, velocity profile, SOC) that are not available in conventional vehicles, providing a wider optimization space. Third, Scenario 3 includes explicit comfort relaxation at low SOC, which produces the largest savings (74 %) through a mechanism not considered in previous studies.

The energy savings are more comparable to those of Nam and Ahn [14], who reported 78.9 % savings for battery thermal management at low temperatures. However, their work addresses battery cooling rather than cabin climate, and the high savings stem from avoiding unnecessary active cooling that a rule-based controller applies. Wang et al. [8] achieved 9 % savings by exploiting speed preview for A/C optimization in connected vehicles, and Amini et al. [9] reported up to 5.4 % savings through hierarchical MPC with robust speed prediction handling. Both values are substantially lower than those achieved

here, confirming that the combination of multiple forecast types (passengers, velocity, SOC) provides a wider optimization space than speed preview alone.

The coordinated recirculation control demonstrated in Scenarios 1 and 3 aligns with the findings of Wei et al. [16], who showed that dynamic recirculation can reduce climate power by 27 % while maintaining acceptable CO_2 levels, and Lesage et al. [17], who demonstrated range gains of 9 km to 26 km through joint optimization of blower and recirculation. The present MPC extends these concepts by embedding recirculation control within a multi-objective optimization that simultaneously considers temperature, CO_2, and energy. The comfort–range trade-off formalized by Lahlou et al. [18] using dynamic programming is here addressed through the SOC stage parameter mechanism, which offers the advantage of online receding-horizon optimization rather than offline trajectory planning.

5.2. Practical Implications for Fleet Operation

The results have direct implications for robotaxi fleet economics. HVAC energy constitutes 30 % to 50 % of total energy consumption during urban driving [1,2]. Reducing HVAC consumption by 58 % to 74 % translates to a meaningful range extension, fewer charging stops, and reduced infrastructure requirements. The SOC-dependent comfort relaxation (Scenario 3) provides an additional safety margin: rather than maintaining full comfort until the battery is depleted, the MPC ensures the vehicle can reach a charger by consciously reducing HVAC load.

The forecast information required by the MPC is inherently available in a robotaxi's fleet management system: booking schedules are known in advance, routes are pre-planned, and SOC trajectories can be predicted from the planned route. No additional sensors or infrastructure are needed beyond what autonomous fleet operation already requires.

5.3. Limitations

Several limitations of this study should be noted. First, all results are obtained from simulation using a lumped-parameter model. While the four-node model captures the dominant thermal dynamics, real vehicles exhibit additional effects such as door opening losses, seat heating, window defogging, compressor cycling, and dehumidification. Second, the PID baseline, while representative of conventional controllers, may underestimate the performance of production-grade climate control systems that incorporate some feedforward logic. Third, the interior thermal mass (T_{mass}) is hidden from the MPC, which may lead to suboptimal behavior during prolonged solar soak conditions where the mass temperature significantly exceeds the cabin air temperature. Fourth, the SOC stage parameter uses a single threshold; a continuous weighting function could provide smoother transitions.

5.4. Future Work

This work establishes the conceptual validity of forecast-exploiting MPC for robotaxi cabin climate control using physics-based prediction models. The deliberate restriction to non-safety-critical cabin conditioning (Section 2.1) enables a natural progression toward data-driven prediction models.

The industry benchmark for intelligent thermal control in robotaxis is set by Zheng and Mueller [3], who described the Zoox thermal system as a tightly coupled MIMO problem with three interacting loops (coolant, refrigerant, cabin air), approximately 100 sensors, and over 100 I/Os. To manage this complexity, Zoox employs machine learning in three supporting roles: a time-series model for pre-trip power budget prediction, adaptive state-space observers for zone-level temperature estimation, and a reinforcement-learning-based system that learns individual passenger comfort preferences over time. However, the core control logic relies on lookup tables and feedforward compensation rather than online optimization. This represents a deliberate trade-off: lookup tables are computationally

inexpensive and deterministic, but they cannot systematically enforce constraints, balance competing objectives, or exploit a receding prediction horizon.

The present work demonstrates that MPC provides precisely these capabilities—constraint handling, multi-objective optimization, and anticipatory control—using physics-based WhiteBox models with a limited number of states. The natural next step is to combine the strengths of both paradigms: *machine-learning-based prediction models within an MPC optimization framework*. This combination would retain the MPC's ability to enforce hard constraints (e.g., $CO_2 \leq 1200$ ppm), optimize multiple actuators simultaneously, and anticipate future disturbances, while leveraging neural networks to handle the modeling complexity that physics-based approaches cannot capture at scale.

Concretely, replacing the WhiteBox predictors with neural networks (ANN or LSTM) trained on real vehicle data would eliminate the need for explicit physical parameter identification and capture effects not modeled in the lumped-parameter approach—door openings, seat heating, dehumidification, compressor cycling, and the inter-loop coupling between cabin, battery, and compute thermal management [6,7]. Nam and Ahn [14] demonstrated the viability of ANN-based MPC for battery thermal management, achieving up to 78.9 % energy savings with reduced computational load compared to physics-based models. A similar approach for cabin climate—where the neural network serves as the MPC's internal prediction model—would combine data-driven accuracy with optimization-based control.

Crucially, robotaxi fleets provide a scalable data pipeline for such models. Fleet vehicles continuously collect cabin temperature, HVAC actuator, and environmental data during operation. By clustering these data by thermal load profile (e.g., summer urban, winter highway) and usage pattern (e.g., high-frequency short trips, airport shuttles), cluster-specific prediction models can be trained that are tailored to the dominant dynamics of each operating regime. A vehicle predominantly serving short urban trips in hot climates would receive a model optimized for rapid cool-down transients and frequent door openings, while a highway shuttle would benefit from a model emphasizing steady-state efficiency and speed-dependent heat exchanger dynamics. These specialized models can be deployed to matching vehicles via over-the-air updates, creating an online learning paradigm where each vehicle benefits not only from its own operational data but from the collective experience of all fleet vehicles operating under similar conditions [19]. This fleet-wide, cluster-based learning approach parallels Zoox's ML pipeline but embeds the learned models within an MPC optimization framework, enabling systematic constraint enforcement and multi-objective optimization that pure ML-based feedforward control cannot provide.

A further data source lies in passenger feedback. Ride-hailing applications routinely collect user ratings and can include targeted questions about thermal comfort. Correlating subjective comfort assessments with objective sensor data would allow the MPC's comfort band and weighting parameters to be refined continuously—similar to the Zoox recommendation system [3], but integrated into the MPC's cost function rather than applied as a separate target adjustment layer.

Beyond passenger vehicles, the forecast-exploiting MPC architecture is transferable to autonomous freight transport. Refrigerated autonomous trucks share the same structural advantage: the route, cargo schedule, and ambient conditions along the trajectory are known in advance. An MPC that anticipates altitude-dependent ambient temperature changes, loading dock stops, and door openings could reduce refrigeration energy while maintaining cold chain integrity.

Real-world validation on an instrumented test vehicle remains essential to quantify the actual energy savings achievable under production conditions and to identify model–

reality gaps that motivate the transition to learned models. Padisala and Dey [15] recently demonstrated MPC-based power allocation between propulsion, HVAC, and battery pre-conditioning for autonomous EVs under cold conditions, confirming that the integrated energy management concept pursued here is gaining traction in the research community. The approach demonstrated by Fussey et al. [13]—adaptive MPC validated on a Jaguar I-PACE on a climatic chassis dynamometer—provides a methodological template for such validation, particularly regarding embedded implementation constraints and real-time solver feasibility. Similarly, the embedded NMPC implementation by Glos et al. [10] on an automotive-grade microcontroller confirms that the computational requirements of cabin climate MPC are compatible with production hardware.

6. Conclusions

This paper demonstrated that model predictive control can exploit the forecast availability unique to autonomous electric vehicles to achieve substantial energy savings in cabin climate management. A four-node lumped-parameter thermal model with CO_2 tracking was developed, and an MPC formulation with four manipulated variables, a dead-band comfort objective, and a novel SOC stage parameter was presented.

Three simulation scenarios, each isolating a distinct MPC capability, yielded the following key findings:

1. Predictive pre-conditioning using booking data achieves 58 % energy savings by operating the heat pump at efficient partial-load conditions rather than maximum power.
2. Anticipation of highway speeds exploits the speed-dependent heat exchanger efficiency (η_{rad}), yielding 60 % savings during the highway phase.
3. SOC-dependent comfort relaxation enables graceful degradation, saving 74 % of energy by consciously trading comfort for range when the battery is critically low.

The underlying principle—that autonomous vehicles possess comprehensive forecast information that predictive controllers can exploit—applies broadly to fleet thermal management and suggests that significant energy efficiency gains are achievable without hardware modifications, purely through intelligent control.

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Abbreviations

The following abbreviations are used in this manuscript:

AEV	Autonomous Electric Vehicle
BMS	Battery Management System
COP	Coefficient of Performance
EV	Electric Vehicle
HP	Heat Pump
HTC	Heat Transfer Coefficient
HVAC	Heating, Ventilation, and Air Conditioning
MPC	Model Predictive Control
MV	Manipulated Variable
NLP	Nonlinear Program
PID	Proportional–Integral–Derivative
PLR	Partial Load Ratio
PTC	Positive Temperature Coefficient (heater)
SOC	State of Charge

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